

DEEP LEARNING LAB – END SEMESTER EXAMINATION (ESE)

Time: 2 Hours

- **Implementation: 1Hr 45 mins**
- **Viva: 15 Minutes**

Instructions

- **Attempt any ONE problem assigned by the examiner.**
- **Use only the provided dataset or approved links.**
- **Display outputs clearly and justify your results during viva.**

EXP 1: MLP (Multilayer Perceptron)

- 1. Design a neural network for digit classification and analyze the effect of changing activation functions.**
- 2. Train neural networks with different numbers of hidden layers and compare their performance.**
- 3. Evaluate how model depth influences convergence and accuracy on a given dataset.**
- 4. Analyze the impact of activation functions on training stability and output predictions.**

EXP 2: CNN

- 1. Develop a convolution-based model for image classification and compare its performance with a fully connected model.**
- 2. Analyze the role of pooling layers in improving model performance and reducing overfitting.**
- 3. Visualize feature maps from intermediate convolution layers and interpret their significance.**
- 4. Study how changing convolution parameters affects classification accuracy.**

EXP 3: Transfer Learning

- 1. Adapt a pretrained model for a new classification task using feature extraction.**
- 2. Compare performance when using frozen layers versus partial fine-tuning.**

3. Analyse how pretrained features improve learning on small datasets.
4. Evaluate training efficiency when only selected layers are updated.

EXP 4: CNN vs Transfer Learning

1. Compare the performance of a model trained from scratch with a pretrained model on the same dataset.
2. Analyse differences in training time and accuracy between the two approaches.
3. Evaluate how the dataset size impacts both methods.
4. Identify scenarios where transfer learning is more beneficial.

EXP 5: Performance Analysis of CNN

1. Study the effect of learning rate on model convergence and stability.
2. Analyse how increasing epochs influence overfitting and generalisation.
3. Plot and interpret training vs validation loss curves.
4. Compare model performance under different hyperparameter settings.

EXP 6: Object Detection

1. Apply a pretrained object detection model to identify objects in images.
2. Analyse confidence scores and detection accuracy for multiple objects.
3. Identify cases where detection fails and explain possible reasons.
4. Evaluate how image quality affects detection performance.

EXP 7: Simple RNN

1. Develop a sequence prediction model for next-step prediction in text data.
2. Analyse the effect of sequence length on prediction accuracy.
3. Study the limitations of basic recurrent models in capturing long-term dependencies.
4. Evaluate model performance on short versus long sequences.

EXP 8: LSTM

1. Implement a model capable of handling long-term dependencies in sequential data.
2. Compare its performance with a simple recurrent model.
3. Analyse the improvement in loss convergence and prediction quality.

4. Evaluate how memory mechanisms affect sequence learning.

EXP 9: GRU

1. Develop a sequence model optimised for faster training.
2. Compare performance and training time with another recurrent architecture.
3. Analyse efficiency and accuracy trade-offs.
4. Evaluate the suitability of the model for real-time applications.

EXP 10: GAN

1. Implement a generative model to create synthetic data samples.
2. Analyse how generated outputs improve over training iterations.
3. Compare generator and discriminator performance during training.
4. Evaluate the quality of generated samples and identify limitations.

Datasets:

1. Fashion-MNIST (subset)
2. MNIST (subset)
3. Cats vs Dogs subset (~1200 images)
4. Custom images (traffic/animals)
5. Text (1000 chars)
6. CIFAR-10 (subset)
7. Custom dataset (medical/animals)
8. Cocomat dataset